

digital

DIGITAL AlphaServer™ Systems

Exceptional performance with a full range of solutions

Powerful, reliable, scalable, flexible – anywhere you look up or down the line, you will find the same commitment to excellence that has propelled DIGITAL AlphaServer systems to a new level of performance. Your productivity will soar with blazing 64-bit power and speed and Internet-Energized software that will drive your expansion into new markets like never before.

Benefits

- A price/performance ratio that's a cut above the rest – with superior performance by any industry standard.

- Advanced clustering technology that ensures maximum uptime regardless of your operating system – resulting in uncompromised data integrity and availability.

- The *only series of solutions with the freedom of choice* of three operating system environments – all AlphaServer systems are designed to run DIGITAL UNIX®, OpenVMS™, or Windows NT™.

- Continuous investment protection lets you buy with confidence. DIGITAL has anticipated and addressed the one constant in your server needs – *change* – through processor upgrades, clustering, and highly scalable and flexible configurations.

- Exceptional systems software is par for the course with DIGITAL ServerWORKS™ Manager, Internet-Energized, and UPS Power Management™ software included with all AlphaServer systems.

- Growth opportunities abound with industry-standard PCI and EISA slots to attach low-cost, industry-standard devices.



advantage

DIGITAL AlphaServer systems ... the obvious choice

Unmatched investment protection

Only DIGITAL delivers an affordable solution that surpasses your highest performance expectations today – and positions you for virtually unlimited growth for years to come. The road to your system's future is clearly marked with migration-free hardware, software, and option upgrades. You can expand seamlessly throughout our entire line of servers – overcoming any obstacles to your business' continued progress.

In addition, powerhouse 64-bit DIGITAL AlphaServer systems are designed for the greatest flexibility in growing your system, allowing you to easily attach low-cost industry standard devices, expand your system's memory, and add greater storage, networking and I/O capabilities as the need arises. The only constant in your server requirements is constant change, and AlphaServer systems are best equipped to meet the challenges you will face.

Exceptional performance with extraordinary value

By any standard, DIGITAL AlphaServer systems lead the field in price/performance benchmarks. Whether you take your measure from industry-standard benchmarks like tpmC®, SPEC®, LINPACK®, SPECweb® – or leading applications such as Microsoft Exchange™, Oracle®, or PeopleSoft® – AlphaServer systems allow you to process more work in less time without breaking your budget. One DIGITAL customer reports building an AlphaServer systems solution that provides them with superior performance at *one-tenth the cost of competing systems*. This is the kind of performance DIGITAL can put to work for you.

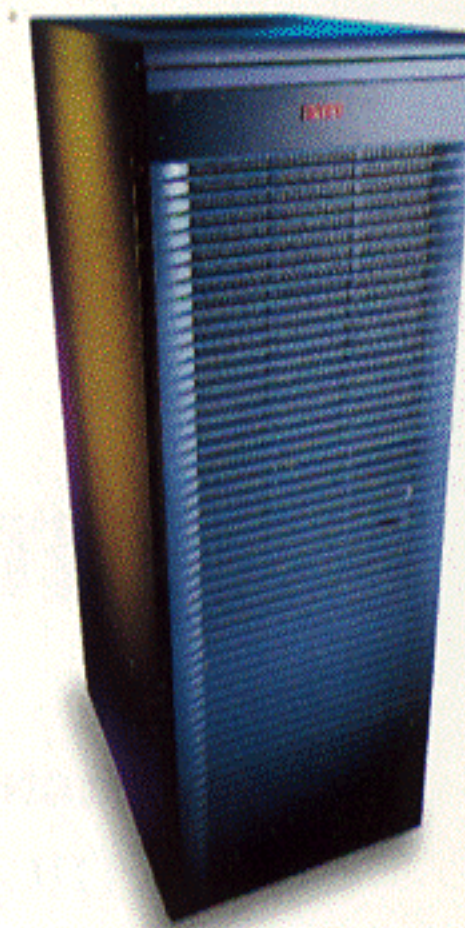
Moreover, AlphaServer systems from DIGITAL are well balanced, and address the issue of capacity far more completely than other systems by including VLM64 (Very Large Memory) capabilities, extremely high storage capacities, fast I/O, and high system bus bandwidth.

Uncompromised availability and data integrity

Maximum uptime is the only acceptable measure in today's competitive business environment. By clustering a series of AlphaServer systems, you reach new levels of performance – with the most advanced clustering technology in the world working for you. If one system goes down, the others seamlessly take over processing – resulting in a higher level of availability than any standalone solution can deliver. When it comes to data availability and integrity, there can be no compromise.



An unprecedented choice of solutions ...



AlphaServer 800 5/333, 5/400 and 5/500 Workgroup Server

AlphaServer 1200 5/400 Workgroup Servers

CPU features	5/333	5/400	5/500	5/400
Number of processors	1	1	1	Up to 2
CPU/clock speed	21164/333 MHz	21164/400 MHz	2164/500 MHz	21164/400 MHz
Cache size on chip	8 KB I, 8 KB D 96 KB secondary	8 KB I, 8 KB D 96 KB secondary	8 KB I, 8 KB D 96 KB secondary/2 MB	8 KB I, 8 KB D 96 KB secondary
Cache size on board	2 MB	2 MB	2 MB	4 MB per processor
In-cabinet CPU upgrade	Yes	Yes	-	Yes
Performance				
tpmC @ \$/tpmC*	-	-	-	-
SPECint95*	10.1	11.7	14.2	12.1
SPECfp95*	12.9	13.7	16.1	17.2
SPECfp95 SMP	-	-	-	24.2 (2 CPU)
SPECint_rate95*	90.9	105	129	211 (2 CPU)
SPECfp_rate95*	116	123	145	275 (2 CPU)
LINPACK 1000 X 1000	374.6	412.5	496.90	1,019 (2 CPU)
SPECweb96*	-	934	1,158	-
Configurations				
Maximum memory	2 GB	2 GB	2 GB	2 GB
Maximum disk storage (in cabinet/total)	72 GB/5.3 TB	72 GB/5.3 TB	72 GB/5.3 TB	Pedestal: 126 GB/7.9 TB
Maximum I/O bandwidth	266 MB/s	266 MB/s	266 MB/s	500 MB/s
I/O support	3 PCI slots, 2 EISA slots, 1 64-bit PCI/EISA slot	3 PCI slots, 2 EISA slots, 1 64-bit PCI/EISA slot	3 PCI slots, 2 EISA slots, 1 64-bit PCI/EISA slot	4 64-bit PCI, 1 32-bit PCI 1 shared 64-bit PCI/EISA 2 PCI channels
Reliability/high-availability features				
OpenVMS Clusters	Ethernet, FDDI, SCSI, DSSI	Ethernet, FDDI, SCSI, DSSI	Ethernet, FDDI, SCSI, DSSI	Ethernet, DSSI, FDDI, SCSI
UNIX Clusters (DIGITAL UNIX)	TruCluster Available Server, TruCluster Production Server, Parallel Software Environment (PCI to Memory Channel Interconnect)			TruCluster Available Server, (PCI to Memory Channel Int
DIGITAL Clusters for Windows NT	Supported	Supported	Supported	Supported
High-availability features	Auto reboot, thermal management, remote system management, disk hot-swap, memory failover, ECC memory, ECC cache, optional uninterruptible power supply (UPS) Also with rackmount: multiple systems, multiple power sources, dual-ported storage			Auto reboot, thermal man memory failover, ECC me error logging, optional uni multiple systems, multiple
I/O options	Ethernet, Fast Ethernet, FDDI, Token Ring, sync. comm., async. comm., RAID, Fast SCSI-2, FW SCSI-2, UltraSCSI, DSSI (OpenVMS), Prestoserve®, ATM			Ethernet, Fast Ethernet, FDI Fast SCSI-2, FW SCSI-2, FW
Software features				
Operating systems	DIGITAL UNIX, OpenVMS, Windows NT			DIGITAL UNIX, OpenVMS, W
Standard software	Internet-Energized, ServerWORKS Manager, UPS Power Management			Internet-Energized, ServerW

Individual workloads should be carefully evaluated before making performance estimates for specific applications. In this chart, no warranty of system performance is expressed or implied. Details on configurations are found in the *DIGITAL Systems and Options Catalog*. Features may differ among operating environments. Performance may vary depending on configuration, application, and operating environment.



5/400 and 5/533

AlphaServer 4000 5/400, 5/466, 5/533 and 5/600 Departmental Servers

5/533	5/400	5/466	5/533	5/600
Up to 2	Up to 2	Up to 2	Up to 2	Up to 2
21164/533 MHz	21164/400 MHz	21164/466 MHz	21164/533 MHz	21164/600 MHz
8 KB I, 8 KB D	8 KB I, 8 KB D	8 KB I, 8 KB D	8 KB I, 8 KB D	8 KB I, 8 KB D
96 KB secondary	96 KB secondary	96 KB secondary	96 KB secondary	96 KB secondary
4 MB per processor	4 MB per processor	4 MB per processor	4 MB per processor	8 MB per processor
Yes	Yes	Yes	Yes	Yes
7,023@\$49	—	—	—	—
16.6	12.1	14.1	16.6	18.8
21.9	17.2	19.2	21.9	29.2
30.5 (2 CPU)	24.2 (2 CPU)	26.5 (2 CPU)	30.5 (2 CPU)	38.9 (2 CPU)
291 (2 CPU)	211 (2 CPU)	250 (2 CPU)	292 (2 CPU)	334 (2 CPU)
342 (2 CPU)	275 (2 CPU)	301 (2 CPU)	342 (2 CPU)	489 (2 CPU)
1,277 (2 CPU)	1,019 (2 CPU)	1,119 (2 CPU)	1,268 (2 CPU)	—
2,258 (2 CPU)	—	1,240 (1 CPU) 1,700 (2 CPU)	—	—
2 GB	4 GB	4 GB	4 GB	4 GB
Pedestal: 126 GB/7.9 TB	Pedestal: 380 GB/over 15 TB Cabinet: 1 TB/over 15 TB	Pedestal: 380 GB/over 15 TB Cabinet: 1 TB/over 15 TB	Pedestal: 380 GB/over 15 TB Cabinet: 1 TB/over 15 TB	Pedestal: 380 GB/over 15 TB Cabinet: 1 TB/over 15 TB
500 MB/s	500 MB/s (1 GB/s with I/O expansion)			
4 64-bit PCI, 1 32-bit PCI 1 shared 64-bit PCI/EISA 2 PCI channels	8-16 64-bit PCI slots (including 3 shared PCI/EISA slots) 2-4 64-bit PCI channels			
Ethernet, DSSI, FDDI, SCSI, CI	Ethernet, DSSI, FDDI, SCSI, CI	Ethernet, DSSI, FDDI, SCSI, CI	Ethernet, DSSI, FDDI, SCSI, CI	Ethernet, DSSI, FDDI, SCSI, CI
TruCluster Production Server, Parallel Software Environment (PCI to Memory Channel Interconnect)	TruCluster Available Server, TruCluster Production Server, Parallel Software Environment (PCI to Memory Channel Interconnect)			
Supported	Supported	Supported	Supported	Supported
management, remote system management, RAID, disk hot-swap, memory, ECC cache, ECC system bus, SMP CPU failover, uninterruptible power supply (UPS). Also with rackmount: multiple power sources, dual-ported storage	Auto reboot, thermal management, remote system management, RAID, disk hot-swap, memory failover, ECC memory, ECC cache, ECC system bus, SMP CPU failover, error logging, optional uninterruptible power supply (UPS). Also with rackmount: multiple systems, multiple power sources, dual-ported storage			
Ethernet, Token Ring, sync. comm., async. comm., RAID, Fast SCSI-2, UltraSCSI, CI, DSSI (OpenVMS), Prestoserve, ATM	Ethernet, Fast Ethernet, FDDI, Token Ring, sync. comm., async. comm., RAID, Fast SCSI-2, FW SCSI-2, FWD SCSI-2, UltraSCSI, CI, DSSI (OpenVMS), Prestoserve, ATM			
Windows NT	DIGITAL UNIX, OpenVMS, Windows NT			
SERVERWORKS Manager, UPS Power Management	Internet-Energized, ServerWORKS Manager, UPS Power Management			



AlphaServer 4100 5/400, 5/466, 5/533 and 5/600 High-End Departmental Servers

	5/400	5/466	5/533	5/600
	Up to 4	Up to 4	Up to 4	Up to 4
	21164/400 MHz	21164/466 MHz	21164/533 MHz	21164/600 MHz
	8 KB I, 8 KB D	8 KB I, 8 KB D	8 KB I, 8 KB D	8 KB I, 8 KB D
	96 KB secondary	96 KB secondary	96 KB secondary	96 KB secondary
	2 MB per processor	4 MB per processor	4 MB per processor	8 MB per processor
	Yes	Yes	Yes	Yes
	—	—	12,971@\$97	15,100@\$80
	12.1	14.1	16.6	18.8
	17.2	19.2	21.9	29.2
	33.4 (4 CPU)	36.1 (4 CPU)	42.1 (4 CPU)	51.4 (4 CPU)
	422 (4 CPU)	485 (4 CPU)	575 (4 CPU)	657 (4 CPU)
	439 (4 CPU)	466 (4 CPU)	565 (4 CPU)	858 (4 CPU)
	1,841 (4 CPU)	1,934 (4 CPU)	2,038 (4 CPU)	2,634 (4 CPU)
	—	2,304 (4 CPU)	3,031 (4 CPU)	3,273 (4 CPU)
	8 GB	8 GB	8 GB	8 GB
15 TB	Pedestal: 380 GB/over 15 TB	Pedestal: 380 GB/over 15 TB	Pedestal: 380 GB/over 15 TB	Pedestal: 380 GB/over 15 TB
TB	Cabinet: 1 TB/over 15 TB	Cabinet: 1 TB/over 15 TB	Cabinet: 1 TB/over 15 TB	Cabinet: 1 TB/over 15 TB
	500 MB/s	500 MB/s	500 MB/s	500 MB/s
	8 64-bit PCI slots (including 3 shared PCI/EISA slots)			
	2 64-bit PCI channels			
SCSI, CI	Ethernet, DSSI, FDDI, SCSI, CI	Ethernet, DSSI, FDDI, SCSI, CI	Ethernet, DSSI, FDDI, SCSI, CI	Ethernet, DSSI, FDDI, SCSI, CI
	TruCluster Available Server, TruCluster Production Server, Parallel Software Environment (PCI to Memory Channel Interconnect)			
	Supported	Supported	Supported	Supported
	Auto reboot, thermal management, disk hot-swap, memory failover, ECC memory, ECC cache, ECC system bus, SMP CPU failover, error logging, optional uninterruptible power supply (UPS). Also with a CAB: multiple systems, multiple power sources, dual-ported storage			
	Ethernet, Fast Ethernet, FDDI, Token Ring, sync. comm., async. comm., RAID, Fast SCSI-2, FW SCSI-2, FWD SCSI-2, UltraSCSI, CI, DSSI (OpenVMS), Prestoserve, ATM			
	DIGITAL UNIX, OpenVMS, Windows NT			
	Internet-Energized, ServerWORKS Manager, UPS Power Management			



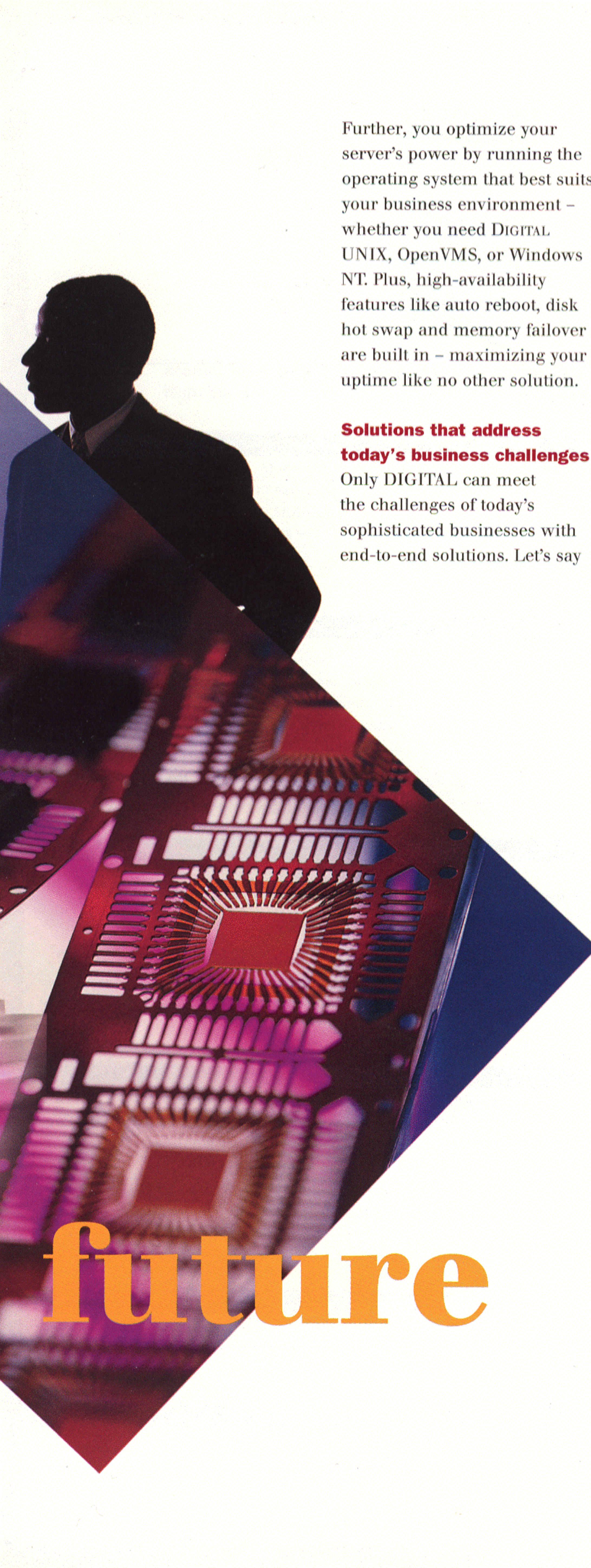
AlphaServer 8200 5/440 and 5/625 Enterprise Servers

AlphaServer 8400 5/440 and 5/625 Enterprise Servers

5/440	5/625	5/440	5/625
Up to 6	Up to 6	Up to 14	Up to 14
21164/437 MHz	21164/612 MHz	21164/437 MHz	21164/612 MHz
8 KB I, 8 KB D	8 KB I, 8 KB D	8 KB I, 8 KB D	8 KB I, 8 KB D
96 KB secondary	96 KB secondary	96 KB secondary	96 KB secondary
4 MB per processor	4 MB per processor	4 MB per processor	4 MB per processor
CPU, I/O, memory, storage	CPU, I/O, memory, storage	CPU, I/O, memory, storage	CPU, I/O, memory, storage
—	—	See note ²	24,537@ \$110
13.6	18.4	13.6	18.4
17.0	20.8	17.0	20.8
34.3 (6 CPU)	48.0 (6 CPU)	42.6 (8 CPU)	56.7 (8 CPU)
701 (6 CPU)	964 (6 CPU)	1,358 (12 CPU)	2,143 (14 CPU)
588 (6 CPU)	—	1,118 (12 CPU)	1,375 (12 CPU)
3,337 ¹ (6 CPU)	4,003 ¹ (6 CPU)	6,654 ¹ (12 CPU), 24,700 ¹ (64 CPU)	7,283 ¹ (12 CPU)
—	—	—	—
12 GB	12 GB	28 GB	28 GB
720 GB/ 85 TB	720 GB/ 85 TB	864 GB/85 TB	864 GB/85 TB
1.2 GB/s	1.2 GB/s	1.2 GB/s	1.2 GB/s
Up to 132 PCI slots, 8 EISA slots	Up to 132 PCI slots, 8 EISA slots	Up to 144 PCI slots, 8 EISA slots, XMI	Up to 144 PCI slots, 8 EISA slots, XMI
Ethernet, DSSI, FDDI, SCSI, CI	Ethernet, DSSI, FDDI, SCSI, CI	Ethernet, DSSI, FDDI, SCSI, CI	Ethernet, DSSI, FDDI, SCSI, CI
TruCluster Available Server, TruCluster Production Server, Parallel Software Environment (PCI to Memory Channel Interconnect)	TruCluster Available Server, TruCluster Production Server, Parallel Software Environment (PCI to Memory Channel Interconnect)	TruCluster Available Server, TruCluster Production Server, Parallel Software Environment (PCI to Memory Channel Interconnect)	TruCluster Available Server, TruCluster Production Server, Parallel Software Environment (PCI to Memory Channel Interconnect)
Supported	Supported	Supported	Supported
Auto reboot, thermal management, optional redundant power system, RAID, disk hot-swap, memory failover, ECC memory, ECC cache, ECC system bus, SMP CPU failover, error logging, optional battery backup, and optional uninterruptible power supply (UPS)	Auto reboot, thermal management, optional redundant power system, RAID, disk hot-swap, memory failover, ECC memory, ECC cache, ECC system bus, SMP CPU failover, error logging, optional battery backup, and optional uninterruptible power supply (UPS)	Auto reboot, thermal management, optional redundant power system, RAID, disk hot-swap, memory failover, ECC memory, ECC cache, ECC system bus, SMP CPU failover, error logging, optional battery backup, and optional uninterruptible power supply (UPS)	Auto reboot, thermal management, optional redundant power system, RAID, disk hot-swap, memory failover, ECC memory, ECC cache, ECC system bus, SMP CPU failover, error logging, optional battery backup, and optional uninterruptible power supply (UPS)
Ethernet, Fast Ethernet, FDDI, Token Ring, sync. comm., async. comm., RAID, FWD SCSI-2, UltraSCSI, CI, DSSI (OpenVMS), Prestoserve, ATM, HiPPI, Fibre Channel, IPI, Mainframe Channel Connect	Ethernet, Fast Ethernet, FDDI, Token Ring, sync. comm., async. comm., RAID, FWD SCSI-2, UltraSCSI, CI, DSSI (OpenVMS), Prestoserve, ATM, HiPPI, Fibre Channel, IPI, Mainframe Channel Connect	Ethernet, Fast Ethernet, FDDI, Token Ring, sync. comm., async. comm., RAID, FWD SCSI-2, UltraSCSI, CI, DSSI (OpenVMS), Prestoserve, ATM, HiPPI, Fibre Channel, IPI, Mainframe Channel Connect	Ethernet, Fast Ethernet, FDDI, Token Ring, sync. comm., async. comm., RAID, FWD SCSI-2, UltraSCSI, CI, DSSI (OpenVMS), Prestoserve, ATM, HiPPI, Fibre Channel, IPI, Mainframe Channel Connect
DIGITAL UNIX, OpenVMS, Windows NT	DIGITAL UNIX, OpenVMS, Windows NT	DIGITAL UNIX, OpenVMS, Windows NT	DIGITAL UNIX, OpenVMS, Windows NT
Internet-Energized, ServerWORKS Manager, UPS Power Management	Internet-Energized, ServerWORKS Manager, UPS Power Management	Internet-Energized, ServerWORKS Manager, UPS Power Management	Internet-Energized, ServerWORKS Manager, UPS Power Management

¹ LINPACK NxN.

² Model 5/350, 50,390 @ \$505 in a TruCluster (32 CPU)



Further, you optimize your server's power by running the operating system that best suits your business environment – whether you need DIGITAL UNIX, OpenVMS, or Windows NT. Plus, high-availability features like auto reboot, disk hot swap and memory failover are built in – maximizing your uptime like no other solution.

Solutions that address today's business challenges

Only DIGITAL can meet the challenges of today's sophisticated businesses with end-to-end solutions. Let's say

you need to move 50,000 users from your old mail system to a Microsoft® Exchange Server™ messaging, groupware, and collaboration system and get them all connected to the Internet. DIGITAL has vast experience doing just this, with more than 900,000 seats successfully sold in less than one year.

Or perhaps you have limited time to get a massive data warehouse in place that will allow information to flow from the shop floor to the warehouse, sales and marketing, and the executive suite. Or maybe you need to integrate Windows NT applications into your current environment without disrupting your current user base. In either case – in any case – DIGITAL has the talent and experience to customize your solution to your unique business needs ... whatever it takes.

A focus on the future with the 64-bit advantage

Data access times *hundreds of times* faster than 32-bit disk-based databases – that's just part of the promise the 64-bit advantage delivers to today's businesses. Overall, 64-bit systems enjoy huge margins in performance gains over any other comparable solution. Everything runs better in a 64-bit environment. Everything.

While the competition is just getting started, DIGITAL already has years of experience bringing the power of 64-bit computing to real-life, realtime business situations. For example, Internet Service Providers (ISPs) and corporate intranet administrators depend on 64-bit technology from DIGITAL to handle the exponential growth of Web users. Moreover, many businesses rely on DIGITAL technology to effectively handle large, memory-resident databases, to process customer queries by accessing legacy databases or data warehouses, and to access and process complicated images, video, and audio information.

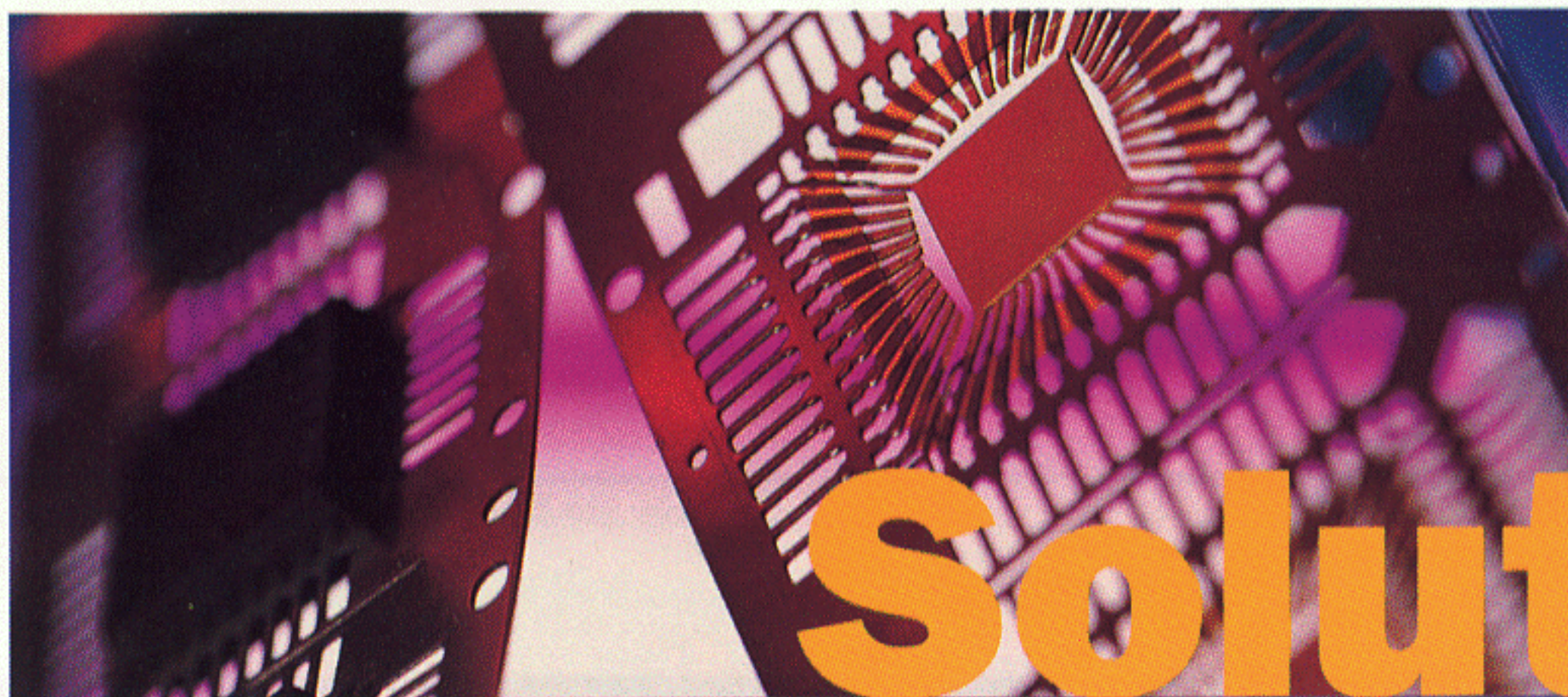
How else is 64-bit power from DIGITAL helping in the real world? One customer reports using VLM64 and a 64-bit database to query live production databases, reduce time from customer order to product shipment, and cut by half the number of mail servers they use worldwide. You may have even experienced 64-bit power firsthand if you have used the AltaVista™ search engine from DIGITAL on the Internet. By searching 30 million Web pages in less than a second, AltaVista produces results other engines just can't match.

Bring the power of 64-bit AlphaServer systems from DIGITAL to your business

DIGITAL provides you with an unprecedented range of choices – allowing you to select the system that best fits your unique business model. When you add up all the advantages, it becomes increasingly obvious that DIGITAL AlphaServer systems are your best choice for an open systems solution. Up and down the line, AlphaServer systems are powerful enough to provide exceptional performance – and scalable enough to ensure that you have a technological edge for years to come. Isn't it about time to bring the 64-bit AlphaServer systems advantage to your business?

future

digital



For more information without complication ...

Explore the DIGITAL AlphaServer systems home page at www.digital.com/alphaserver. Here, you can access a wealth of information on the powerful 64-bit, Internet-Energized DIGITAL AlphaServer systems – from specific **products** and **solutions** to **services**, **promotions** and more.

You can also register to receive the *AlphaServer Update*, an e-mail newsletter featuring the latest news and special offers.

Call us for answers to your questions. You have carte blanche when it comes to calling for our attention.

Whether you need consultation on DIGITAL AlphaServer systems – or would like to inquire about other products or services – please contact your local DIGITAL representative or authorized Business Partner. For information by fax, call 800-DIGITAL via a touch-tone phone in the U.S. or Canada – or 1-908-885-6426 outside the U.S. and Canada.

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PeopleSoft is a trademark of PeopleSoft, Inc.

Strategic Power Management is a trademark of Exide Electronics.

The PCI to Memory Channel Interconnect includes technology from Encore Computer Corporation.